1.5 Chemical Energetics

Question Paper

Course	CIE A Level Chemistry
Section	1. Physical Chemistry
Topic	1.5 Chemical Energetics
Difficulty	Hard

Time allowed: 20

Score: /10

Percentage: /100

Question 1

In the gas phase, phosphorus pentachloride can be thermally decomposed into gaseous phosphorus trichloride and chlorine.

$$PCl_5 \rightarrow PCl_3 + Cl_2$$

The table below gives the relevant bond energies found in these compounds.

Bond	Bond energy / kJ mol^{-1}
P – Cl (in both chlorides)	328
Cl – Cl	241

What is the enthalpy change in the decomposition of the reaction?

A. – 415 kJ mol^{-1}

B. + 415 kJ mol^{-1}

C. + 95 kJ mol^{-1}

D. – 95 kJ mol^{-1}

[1 mark]

Question 2

The equations below show the formation of sulfur oxides from sulfur and oxygen.

$$S\,(s) + O_2\,(g) \rightarrow SO_2\,(g) \quad \Delta H_f^{\ominus} = -297 \text{ kJ mol}^{-1}$$

$$S\,(s) + 1\tfrac{1}{2}\,O_2\,(g) \rightarrow SO_3\,(g) \quad \Delta H_f^{\ominus} = -395 \text{ kJ mol}^{-1}$$

What is the enthalpy change of reaction, $\Delta H_r^{\ominus}$, of $2SO_2\,(g) + O_2\,(g) \rightarrow 2SO_3\,(g)$?

A. + 98 kJ mol^{-1}

B. - 98 kJ mol^{-1}

C. + 196 kJ mol^{-1}

D. - 196 kJ mol^{-1}

[1 mark]

Question 3

In a calorimetric experiment 2.50 g of a fuel is burnt in oxygen. 30 % of the energy released during the combustion is absorbed by 500 g of water, the temperature of which rises from 25 °C to 68 °C.

The specific heat capacity of water is 4.2 J g^{-1} K^{-1}.

What is the total energy released per gram of fuel burnt?

A. 25 284 J

B. 63 210 J

C. 120 400 J

D. 301 000 J

[1 mark]

Question 4

Which equation correctly shows how the bond energy for the covalent bond Y-Z can be calculated by dividing ΔH by n?

A. $n\,YZ(g) \rightarrow n\,Y(g) + \frac{n}{2}\,Z_2(g)$

B. $Z(g) + YZ_{n-1}(g) \rightarrow YZ_n(g)$

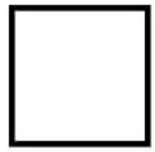
C. $2\,YZ_n(g) \rightarrow 2\,YZ_{n-1}(g) + Y_2(g)$

D. $YZ_n(g) \rightarrow Y(g) + nZ(g)$

[1 mark]

Question 5

The diagram shows the skeletal formula of cyclobutane.



The enthalpy change of formation of cyclobutane is +75.1 kJ mol^{-1}, and the enthalpy change of atomisation of graphite is +712 kJ mol^{-1}.

The bond enthalpy of C–H is 390 kJ mol^{-1} and of H–H is 429 kJ mol^{-1}.

What is the average bond enthalpy of the C–C bond in cyclobutane, rounded to the nearest whole number?

A. 236 kJ mol^{-1}

B. 315 kJ mol^{-1}

C. 342 kJ mol^{-1}

D. 700 kJ mol^{-1}

[1 mark]

Question 6

Some bond energy values are listed below.

Bond	Bond energy / kJ mol^{-1}
Br – Br	194
Cl – Cl	247
C – H	412
C – Cl	338

These bond energy values relate to the following four reactions.

W	$Br_2 \rightarrow 2Br$
X	$2Cl \rightarrow Cl_2$
Y	$CH_3 + Cl \rightarrow CH_3Cl$
Z	$CH_4 \rightarrow CH_3 + H$

What is the correct order of enthalpy changes of the above reactions from most negative to most positive?

A. Y → Z → W → X

B. Z → W → X → Y

C. Y → X → W → Z

D. X → Y → Z → W

[1 mark]

Question 7

A student calculated the standard enthalpy change of formation of propane, C_3H_8, using a method based on standard enthalpy changes of combustion.

He used correct values for the standard enthalpy change of combustion of propane (–2220 kJ mol^{-1}) and hydrogen (–286 kJ mol^{-1}) but he used an incorrect value for the standard enthalpy change of combustion of carbon. He then performed his calculation correctly. His final answer was –158 kJ mol^{-1}.

What did he use for the standard enthalpy change of combustion of carbon?

A. -1234 kJ mol^{-1}

B. -411.3 kJ mol^{-1}

C. -2084 kJ mol^{-1}

D. -694.7 kJ mol^{-1}

[1 mark]

Question 8

Given the following enthalpy changes,

$I_2(s) \rightarrow I_2(g)$ $\quad \Delta H^{\ominus} = +38$ kJ mol^{-1}

$I_2(g) + 3Cl_2(g) \rightarrow 2ICl_3(s)$ $\quad \Delta H^{\ominus} = -214$ kJ mol^{-1}

What is the correct value for $\Delta H_f^{\ominus}$ of iodine trichloride, ICl_3 (s)?

A. –214 kJ mol^{-1}

B. –176 kJ mol^{-1}

C. –88 kJ mol^{-1}

D. +176 kJ mol^{-1}

[1 mark]

Question 9

Using the following information:

$$CO\,(g) + \frac{1}{2}\,O_2\,(g) \rightarrow CO_2\,(g) \qquad \Delta H^{\ominus} = -283\ \text{kJ mol}^{-1}$$

$$H_2\,(g) + \frac{1}{2}\,O_2\,(g) \rightarrow H_2O\,(l) \qquad \Delta H^{\ominus} = -286\ \text{kJ mol}^{-1}$$

$$H_2O\,(g) \rightarrow H_2O\,(l) \qquad \Delta H^{\ominus} = -44\ \text{kJ mol}^{-1}$$

What is the enthalpy change, $\Delta H^{\ominus}$, for the following reaction?

$$CO_2\,(g) + H_2\,(g) \rightarrow CO\,(g) + H_2O\,(g)$$

A. -41 kJ mol^{-1}

B. +41 kJ mol^{-1}

C. -525 kJ mol^{-1}

D. +525 kJ mol^{-1}

[1 mark]

Question 10

The conversion of graphite into diamond is an endothermic reaction.

$$(\Delta H = +3\ \text{kJ mol}^{-1}).$$

$$C\,(\text{graphite}) \rightarrow C\,(\text{diamond})$$

1 The enthalpy change of atomisation of diamond is smaller than that of graphite.

2 The bond energy of the C–C bonds in graphite is greater than that in diamond.

3 The enthalpy change of combustion of diamond is greater than that of graphite.

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

D. 1 only

[1 mark]